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CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 3.01

Indices, Surds & Polynomials

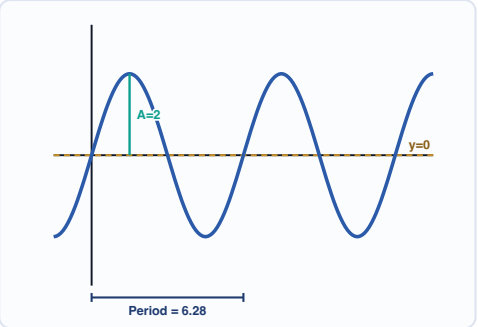
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Indices, Surds & Polynomials Practice Exercises

1 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



A 57 units per second

B 19 units per second

C $\frac{19}{3}$ units per second

D 21 units per second

2 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x - 2)$.

A -10

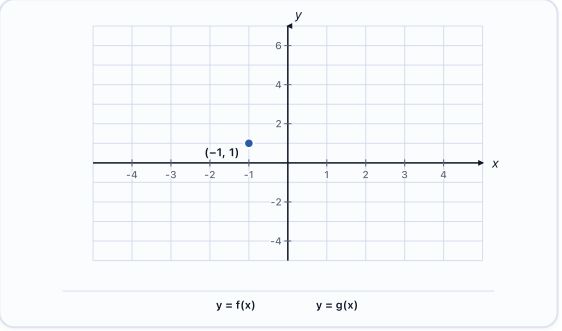
B -7

C -12

D -22

3 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 27

B 36

C 54

D 72

4 MCQ

Solve the logarithmic equation $\log_2(5x+6) - \log_2(x) = 3$.

A $x = 2$

B $x = 6$

C $x = \frac{1}{2}$

D $x = -3$

5 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_3 = 10 \times {}^{n}\mathrm{C}_2$$

A $n = 7$

B $n = 12$

C $n = 5$

D $n = 8$

6 MCQ

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{9}{2}$

D 0

7 MCQ

In $\triangle ABC$, sides of lengths 32 and 10 meet at an included angle of 60° opposite side x. Find the squared length $f = x^2$ by the Law of Cosines.

A 804

B 803

C 820

D 809

8 MCQ

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

A $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$

B $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$

C $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$

D $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

9 MCQ

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants.
Find the remainder when $P(x)$ is divided by $x - 3$.

A 12

B -150

C -30

D 60

10 MCQ

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a > 0$.
A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$.
Find the remainder when $P(x)$ is divided by $(x+a)$.

A -48
B -112
C 48
D 0

11 MCQ

When the polynomial $\mathrm{P}(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x-3$, the remainder is 10. What is the value of the constant a?

A 3
B -9
C -33
D -3

12 MCQ

In the expansion of $(1+ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

A -240
B -160
C 160
D -40

13 MCQ

An arithmetic progression (h_u) has first term $h_1 = 24$ and common difference $d = 17$. Find the term h_5.

A 116
B 109
C 75
D 92

14 MCQ

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x-2$. Find the value of the constant a.

A -3
B -6
C 3
D 0

15 MCQ

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x-4)$. What is the value of the constant a?

A -10.00
B 7.00
C -7.00
D -5.00

16 MCQ

In this problem, the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 5)$. What is the value of the constant a?

A 9.20

B -9.20

C -12.20

D -7.20

17 MCQ

Observe that the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 6)$. What is the value of the constant a?

A -11.33

B -9.33

C -14.33

D 11.33

18 MCQ

Suppose the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 7)$. What is the value of the constant a?

A 13.43

B -13.43

C -11.43

D -16.43

19 MCQ

Determine the solution to the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{7 \ln 5}{\ln 5 - \ln 2}$

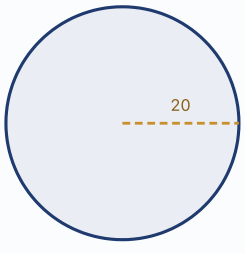
B $x = \frac{7 \ln 2}{\ln 5 - \ln 2}$

C $x = \frac{7}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$

20 MCQ

A sector of a circle of radius $r = 20$ cm has central angle $\theta = 1.5$ rad. What is the area of the sector?



A $300.0\ \text{cm}^2$

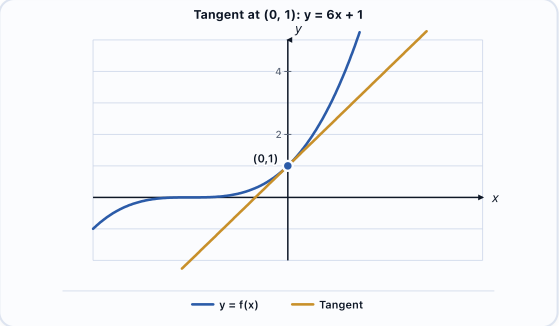
B $30.0\ \text{cm}^2$

C $600.0\ \text{cm}^2$

D $305.0\ \text{cm}^2$

21 MCQ

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.



A $y = 6x + 1$

B $y = 2x + 1$

C $y = 6x - 1$

D $y = 6x + 3$

22 MCQ

Determine the numerical value of the definite integral: $\int_0^1 (3x + 1)^3 \, dx$.

A $\frac{255}{12}$

B $\frac{255}{4}$

C $\frac{255}{3}$

D $\frac{259}{12}$

23 MCQ

In this problem, the function $f(x) = \frac{4x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x-1}{x-4}$

B $f^{-1}(x) = \frac{2x+1}{x-4} \quad (x \neq 4)$

C $f^{-1}(x) = \frac{x-2}{4x+1}$

D $f^{-1}(x) = \frac{4x-2}{x+1}$

24 MCQ

In the expansion of $(1 + \frac{x}{2})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

A 11

B 8

C 5

D 9

25 MCQ

Considering the expansion of $(1 + \frac{x}{3})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

A 12

B 13

C 7

D 15